

ARPA-H Proposal Abstract: Perioperative Daraxonrasib with an LLM-Advised Eight-Arm Robotic Whipple in KRAS-Mutated PDAC

36-month program.

Three gates, each with a stated kill criterion. No gate is a review of intent.

One performer.

An independent scientist with fourteen prior works, contracting one clinical site.

Transition.

An RP2D, a public safety set, and an open verification package usable by any site.

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Intended Partner Site: UC San Diego Moores Cancer Center (feasibility stage)

*Submitted as an independent scientist under the funding approach set out in Science: A New Golden Age (July 2026).
Not medical or regulatory advice, and not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.
UC San Diego is named as the intended partner of choice and is not a sponsor, site, or endorser.*

1 One Performer, Bounded Milestones

Science: A New Golden Age asks the federal research portfolio, roughly **\$200 billion** a year, to stop deferring to “the same incumbents that consume the funding” and to invest directly in individual researchers [1]. Chapter II names the instrument it trusts for high-risk work: the ARPA program-manager model, with ARPA-H and ARPA-E as the live examples, chosen because a program manager can start a bet, run it on milestones, and end it without a committee.

This proposal is written to be run that way. It is one performer, not a consortium. Every milestone below has a stated kill criterion, and each is checkable from data the program itself produces rather than from a narrative progress report. The work that would normally consume the first eighteen months of an ARPA program, the protocol, the regulatory package, and the modeling, is already finished and published [2].

ARPA practice the report endorses	How this program is written for it
A program manager who can end a bet	Three gates, each with a numeric kill, not a narrative review
Milestones tied to measurable outputs	Stop latency in ms, conversion counts, dose-limiting toxicity
Performers outside the incumbent set	One independent scientist contracting a single clinical site
A defined transition out of the program	An RP2D, a public safety set, and open verification package

Four features of the ARPA model that the report endorses, and the specific way each is written into this program so that it can be checked at a gate rather than argued at a review.

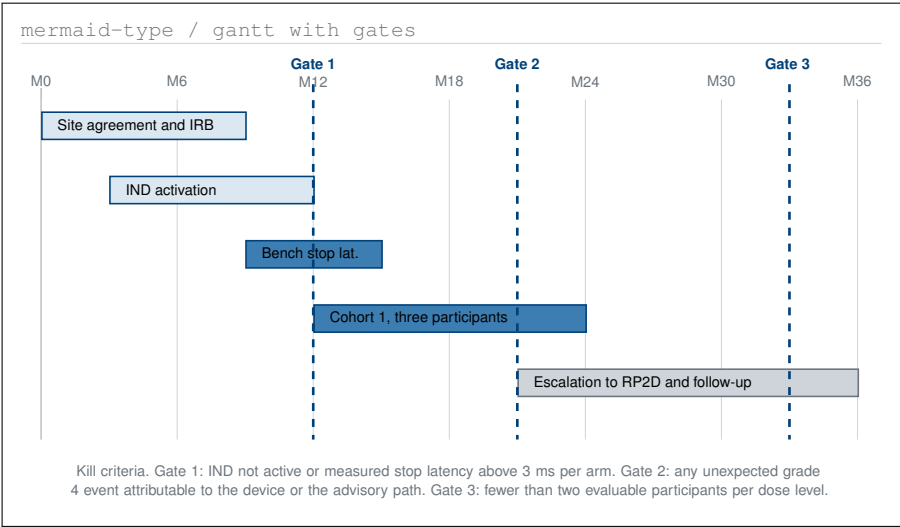
2 Thirty-Six Months, Three Gates

The program is scoped to thirty-six months rather than five years, because an ARPA bet should either produce a recommended Phase 2 dose and a clean safety set or be stopped. Gate 1 closes the regulatory and bench work, Gate 2 closes the first treated cohort, and Gate 3 closes escalation. A gate that fails ends the program; it does not trigger a replan.

3 What Is Already De-Risked

An ARPA program should not spend its first year discovering whether the hypothesis is worth testing. Three questions that would normally occupy that year are answered and published.

Is the drug worth the operation? An August 2025 quantitative systems pharmacology simulation, ten arms and 250 ordinary differential equations per patient, returned median overall survival of **12.8 months** for the daraxonrasib combination against **5.4 months** for chemotherapy [3]. In May 2026 RASolute 302 reported



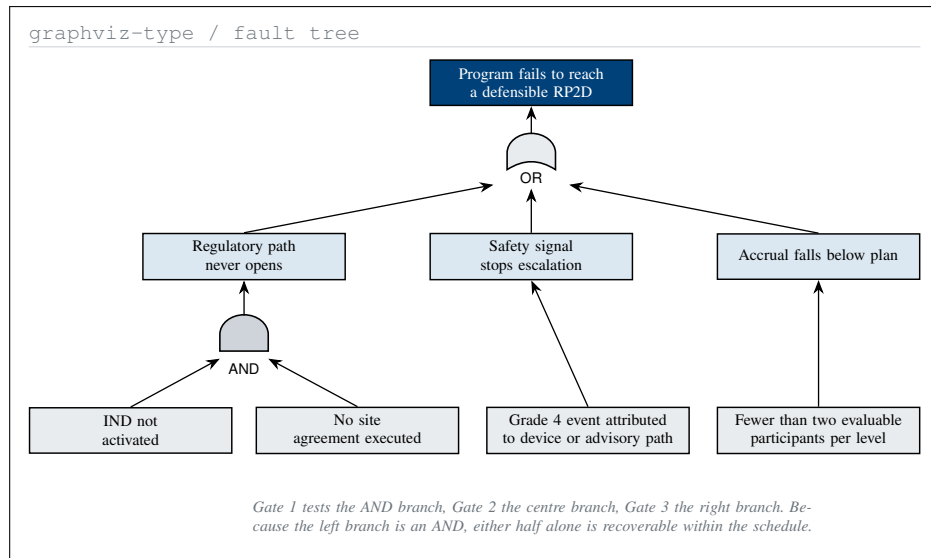
The thirty-six month schedule with its three gates. Each gate carries a numeric kill criterion, so the program manager can end the bet on data rather than on a narrative review.

13.2 against **6.6 months** in the RAS G12 population [4]. The proximity is chronology, not validation: 1000 simulated against 241 enrolled, a combination against a single agent, and KRAS G12C against a primarily G12D and G12V population.

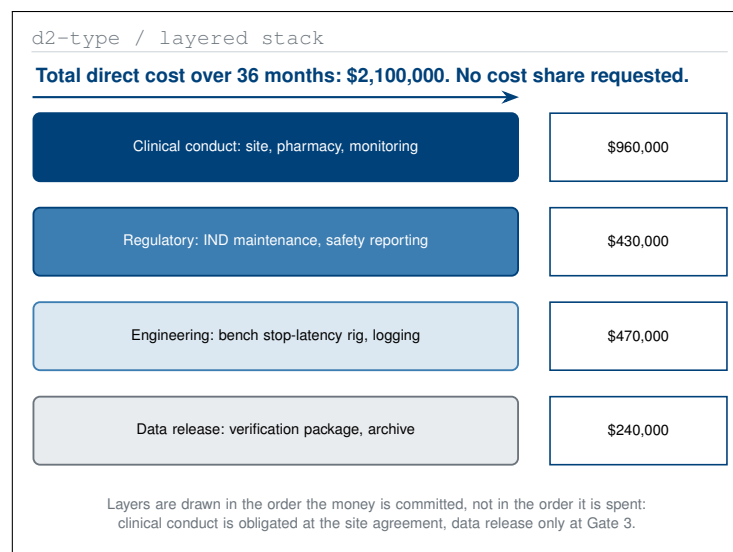
Can the modeling be audited? The digital-twin work carries an ASME V&V 40 and ICH M15 aligned credibility score of **81.9** across 55 verification notebook tests [5, 6, 7].

De-risked item	Result	Why it matters at a gate
Arm selection	mOS HR 0.25, best arm	Gate 2 does not have to re-derive the regimen
Model credibility	Score 81.9, 55 tests	Gate 1 inherits audited model rather than a new one
Regulatory package	IND, protocol published	Gate 1 is an activation check, not an authoring task
Cost per simulation	\$36,330 vs. \$120,000 to \$2,000,000 benchmark	The performer’s burn rate is already demonstrated

Four items retired before any federal dollar was requested, and the gate each one removes work from. Every figure is drawn from the author’s published simulation and regulatory sources.



Every path to program failure, and the gate that tests it. The left branch is an AND gate, so a single missing item there is recoverable inside the thirty-six month schedule.



Where every dollar of the request lands, layer by layer. The layers are ordered by when the money is committed rather than by when it is spent, which is what a gate decision actually turns on.

Gate	Month	Kill criterion, checked from program data
Gate 1	12	IND not active / measured arm-level stop latency > 3ms / system-wide stop > 500ms
Gate 2	21	Any unexpected grade 4 event attributed to device or advisory path in cohort 1
Gate 3	33	Fewer than two evaluable participants at any dose level, or no dose level tolerable

The three kill criteria in full. Each is a number the program itself produces, so the decision to continue does not depend on the performer's own assessment of progress.

Scope of Claims

Claimed: a published protocol and Investigational New Drug application, three audited simulations, and a demonstrated cost per simulation roughly one thirtieth of the industry benchmark. Not claimed: that the simulations validate RASolute 302, that the advisory system holds any regulatory clearance, that anything here has been used in a human, or that UC San Diego has agreed to anything. The trial does not exist until a site, an IRB, an active IND, monitoring, insurance, and a sponsor of record are all in place.

Availability and Conflicts

Every work cited to a DOI is openly deposited; code, verification notebooks, and figure sources are in the repository [12]. The applicant holds no equity in Revolution Medicines or any robotic surgery vendor, and no sponsor, contract research organization, or medical society reviewed this document.

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